Convexity Adjustment in LIBOR in Arrears Swaps

The fixed rate that we solved for in Figure 6.19 is not exactly equal to the fixed rate that would actually be paid or received in a LIBOR in arrears swap. The reason is quite subtle and has to do with the fact that LIBOR rates are being set in an unconventional manner. Remember that in a plain vanilla swap LIBOR is set at the beginning of each period, whereas in a LIBOR in arrears swap, LIBOR is set at the end of each period. This difference actually implies an adjustment to the fixed rate above and beyond that which we saw when we modeled up the LIBOR in arrears swap in Figure 6.19. The nature of this adjustment is referred to as a *convexity adjustment*.

The easiest way to show that a convexity adjustment comes into play when we are pricing a LIBOR in arrears swap is to work through a simple example. Assume we have annual payments on both sides of the swap and suppose that when we bootstrap the swap curve, we come up with the following forward rates:

$$f(0,1) = 5\%, f(1,2) = 6\%, \text{and } f(2,3) = 7\%.$$

That is, today's 1-year LIBOR setting is 5%; 1-year LIBOR, one year forward is 6%; and 1-year LIBOR, two years forward is 7%.

Now let's look at pricing up a 1-year spot starting LIBOR in arrears swap (which we will denote as a 0x1 LIBOR in arrears swap). This swap will have a single LIBOR setting that will occur one year from today. Now, we could go ahead and model up this arrears swap in a similar manner as we did in Figure 6.19, but we can probably come up with the fixed rate for this one in our head. The model would come up with a fixed rate of 6%: since there is only one setting in this LIBOR in arrears swap that the forwards say will be 6%, the fixed rate the model would give us would also be equal to 6%.

But it turns out that the actual fixed rate in this swap cannot be 6%, and the reason for this is due to this so-called convexity adjustment. To show that the fixed rate in this 0x1 LIBOR in arrears swap cannot be 6%, let's assume for the moment that it is, and we will show that leads to an arbitrage. Suppose that we are paying fixed in the 0x1 LIBOR in arrears swap at 6% and let's look at hedging this position using regular swaps. Suppose that we hedge the LIBOR in arrears swap by receiving in a 1x2 plain vanilla swap. This is a forward starting swap that has an effective date in one year, and payments are made two years from today. Now, we can also go ahead and use our model and solve for the fixed rate in this 1x2 swap, but hopefully it is pretty clear that it has to be 6%. If we model up this forward starting swap, there will be one floating payment based on the forward of 6%, and thus the fixed rate that we will solve for will also be 6%.

Note that the LIBOR setting in each swap will be identical; both the 0x1 LIBOR in arrears swap and the 1x2 swap will have LIBOR settings one year from now. A big difference between the two swaps, however, is when cash flows are received: the 0x1 LIBOR in arrears swap pays its cash flows in one year, whereas the 1x2 swap pays cash flows in two years. If we enter into \$100mm of the 0x1 arrears swap, we will need to enter into \$100mm $\times$ 1.06 = \$106mm of the 1x2 swap in order for the two swaps to have the same present value for each dollar of cash flow generated at their respective maturities.[39]

So suppose we pay fixed on \$100mm of the 0x1 arrears swap at the presumed rate of 6% and receive fixed on \$106mm of the 1x2 forward starting swap at the known rate of 6%. Let's assume that we hold each of these trades in our book for one year. We are going to take a look at the value of each of these trades in one year as a function of where 1-year LIBOR is in a year. Let's consider the following three cases:

<u>Case 1</u>: One year has gone by, and 1-year LIBOR has set at 5%.

Let's look at the value of each of the two trades now that a year has gone by. With 1-year LIBOR at 5%, we will be receiving 5% in the 0x1 arrears swap but paying out the fixed rate of 6%. These coupon payments will be made now. So the net cash flow from the arrears swap is

$$\$100\text{mm} \ \times (5\% - 6\%) = \$100\text{mm} \ \times (-1\%) = -\$1\text{mm}. \qquad (6.6)$$

Now let's take a look at the value of the 1x2 forward starting swap, the hedge we put on against this trade. With 1-year LIBOR at 5%, we will be receiving 6% and paying 5% on a notional of \$106mm. Thus, the net coupon payments from this swap will be

$$\$106\text{mm} \ \times (6\% - 5\%). \qquad (6.7)$$

The important thing to realize is that these coupon payments will be received one year from now (i.e., two years from when the trade was put on). Nonetheless, we can value the known cash flows that this swap will pay one year from now. All we need to do is discount the cash flow that will be received in a year. Since it is assumed that 1-year LIBOR is 5% now, this

[39] Remember that if $f(t, t+1)$ denotes the forward LIBOR rate between times t and $t+1$, then \$1 to be received at time t is worth the same as \$1 $\times [1 + f(t, t+1)]$ at time $t+1$. In the example at hand, f(1,2) = 6%, and thus \$1 received at year 1 has the same value as \$1.06 at year 2.

means that the 1-year zero coupon rate must be 5% as well, and thus the present value of this cash flow at the end of the first year is[40]

$$\$106\text{mm} \times (6\% - 5\%)/1.05 = \$100\text{mm} \times (1\%) \times 1.06/1.05. \tag{6.8}$$

Note that the cash flow in Equation 6.8 is greater than \$1mm. The net cash flow generated from this strategy (i.e., paying fixed in the 0x1 arrears swap and receiving fixed on the 1x2 forward swap) is the sum of the cash flow in Equation 6.6 and Equation 6.8, which is a positive quantity.

Case 2: One year has gone by, and 1-year LIBOR has set at 6%.

This case is not too interesting. Since the fixed rate in each swap is 6% and LIBOR has set at 6%, the value of each trade is zero.

Case 3: One year has gone by, and 1-year LIBOR has set at 7%.

Let's look at the value of the two trades again. The 1-year LIBOR rate is at 7%, and we will be receiving 7% in the 0x1 arrears swap but paying out the fixed rate of 6%. So the net value of the arrears swap is

$$\$100\text{mm} \times (7\% - 6\%) = \$100\text{mm} \times (+1\%) = +\$1\text{mm}. \tag{6.9}$$

Let's look at the value of the 1x2 forward starting swap. With 1-year LIBOR at 7%, we will be receiving 6% and paying 7% on a notional of \$106mm. Thus, the net coupon payments from this swap will be

$$\$106\text{mm} \times (6\% - 7\%). \tag{6.10}$$

These coupon payments will be made in one year from now (i.e., two years from when the trade was put on). Since it is assumed that 1-year LIBOR is 7% now, this means that the 1-year zero coupon rate must be 7% as well, and thus the present value of this cash flow at the end of the first year is

$$\$106\text{mm} \times (6\% - 7\%)/1.07 = \$100\text{mm} \times (-1\%) \times 1.06/1.07. \tag{6.11}$$

Note that the cash flow in Equation 6.11 is negative, but its absolute value is less than \$1mm. The net cash flow generated from this strategy is the sum of the cash flow in Equation 6.9 and Equation 6.11, which is a positive quantity.

The results of these three cases can be found in Table 6.8. If LIBOR turns out to be 6% in a year, we break even. But if it is 5% or 7%, we make

[40] To see why the 1-year zero coupon rate must be 5%, go back and take a look at the bootstrapping methodology we went over in Section 3.2.

Assumed Value of 1-year LIBOR in One Year	Value of 0x1 Arrears Swap	Value of 1x2 Forward Starting Swap	Net Value
5%	\$100mm × (5% − 6%) = \$100mm × (−1%)	\$106mm × (6% − 5%)/1.05 = \$100mm × (+1%) × 1.06/1.05	> 0
6%	0	0	= 0
7%	\$100mm × (7% − 6%) = \$100mm × (+1%)	\$106mm × (6% − 7%)/1.07 = \$100mm × (−1%) × 1.06/1.07	> 0

Table 6.8: Value of Arrears Swap and Hedge in One Year Given Different Values of 1-Year LIBOR

money. Moreover, it is pretty easy to convince ourselves that if LIBOR takes on any value other than 6% in a year, then we make money. To this end, denote by $L(1,2)$ the level of LIBOR that we observe in a year. In a year, the net present value of the cash flow from these two swaps is

$$\$100\text{mm} \ \times \left\{ [L(1,2) - 6\%] + \frac{1.06}{1 + L(1,2)} [6\% - L(1,2)] \right\}. \qquad (6.12)$$

Figure 6.21 shows this net present value as a function of the LIBOR rate $L(1,2)$ observed in a year. As this figure shows, the net present value is always nonnegative and strictly positive whenever LIBOR does not set at 6% in a year.

But this, then, would be an arbitrage opportunity: if we could pay in the 0x1 arrears swap at 6% and receive in the 1x2 forward starting swap at 6%, we would have zero cash flows today and would be assured of always having nonnegative cash flows in a year; and in fact we would earn positive cash flows anytime 1-year LIBOR was not exactly 6%. Under the assumption of no arbitrage, this means that the fixed rate in the arrears swap cannot be 6%. The fixed rate must be higher to allow for the possibility of losing money in some scenarios.

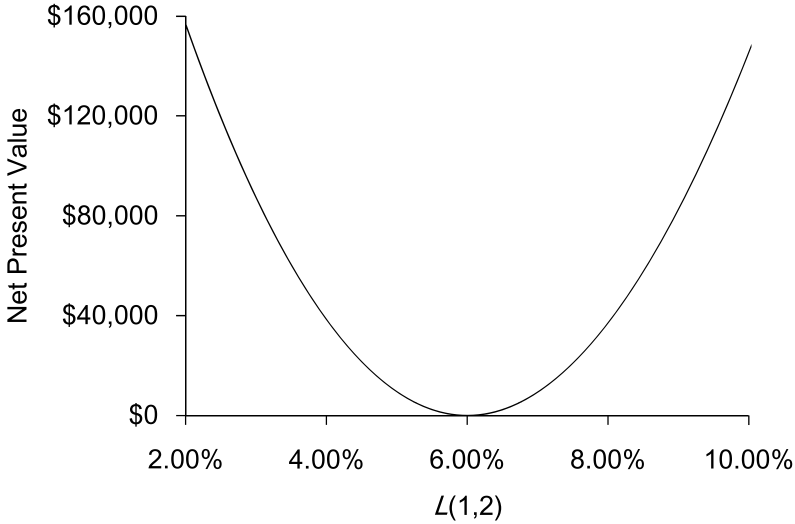

Figure 6.21: Net Present Value of Cash Flows in Arrears Swap and Hedge as a Function of LIBOR in a Year

How much higher should the fixed rate be? That is beyond the scope of this discussion.[41] However, we can make one observation. Most models will place greater value on the payoff profile in Figure 6.21, the higher the implied volatility of the LIBOR forward in one year. This means that the higher the implied vol, the higher the fixed rate will be. In other words, the greater the level of base vol for this forward, the greater the convexity adjustment.

The profile exhibited in Figure 6.21 is somewhat reminiscent of a *cap/floor straddle*, in particular a position consisting of caplet and a floorlet both struck at 6%. This suggests that in order to hedge the exposure to vol, we may want to sell some options (perhaps, heuristically, some caps and floors on LIBOR that expire in a year). Selling these options would help flatten out the P&L profile in Figure 6.21.

To put this another way, the dealer who pays fixed in the LIBOR in arrears swap gets long the option-like profile in Figure 6.21. This profile is a benefit to him. Heuristically, we can think of the convexity adjustment in the trade (i.e., the increase in the fixed rate) as compensation for the options that the dealer implicitly gets long in the trade.

[41] See Brigo and Mercurio (2006) for a further discussion of the pricing of the convexity adjustment in an arrears swap.

6.5.2 CMS Swaps

CMS swaps are swaps in which at least one leg is floating tied to a CMS rate. As we discussed in Chapter 1, a CMS rate is a floating rate of interest. For example, 10-year CMS represents a floating rate of interest reset daily equal to the 10-year swap rate.

Consider the trade in Table 6.9. This is a 1-year trade. It has a single payment made on each side at the end of the year. Notice that the floating side is set equal to the then-prevailing 3-year swap rate, as determined in one year. The payment made in one year by the client is equal to

$$\$100\text{mm} \times 3\text{-year CMS} \times 360/360.$$

Every 1 bp change in 3-year CMS one year from now results in a $10,000 (i.e., 1 bp of $100mm) change in the payment made by the client. We will refer to this trade as the CMS swap.

Let's consider again our stylized example in which swaps pay annual 30/360 on both sides. But in this section we are going to restrict ourselves to the case in which the swap curve is flat, and any change in rates occur only through parallel shifts. In particular, assume that when this trade is first put on the swap curve is as given in Table 6.10.

Dealer Pays:	Fixed
Client Pays:	3-year CMS
Notional:	$100mm
CMS Reset Date:	1 year
Dealer Payment Amount:	Notional × Fixed × 360/360
Client Payment Amount:	Notional × 3-year CMS × 360/360
Payment Date:	1 year

Table 6.9: Terms of a CMS Swap

Maturity	Swap Rate (Ann 30/360)
1	5.00%
2	5.00%
3	5.00%
4	5.00%

Table 6.10: Today's Assumed Swap Curve Used to Examine the CMS Trade

The nice thing about dealing with a flat swap curve is that all zero coupon rates, forward LIBOR rates, and forward swap rates are the same. In particular, in the case of the curve in Table 6.10, it is easy to verify that the 3-year swap rate, one year forward is 5%.

Since the 3-year swap rate, one year forward is 5%, it would seem very reasonable to assume that the appropriate fixed rate in the CMS swap is 5%. That is, the forwards allow us to lock in a 3-year swap rate, one year forward of 5%. Thus, it seems sensible that the fixed rate in this trade would be 5% since the floating rate in this trade references the 3-year swap rate in one year. We will go ahead and make this assumption.

Let's view this trade from the position of the dealer who is paying fixed (and receiving 3-year CMS). Suppose the dealer wants to hedge out some of the risk in this trade by receiving fixed in a plain vanilla swap. In particular, the dealer chooses to receive fixed in a 3-year swap, one year forward (which we will refer to as the "hedge"). There is a particular reason that this swap is being chosen as the hedge. In a year, the value of this swap will depend on where the 3-year swap rate is trading at the time. And in a year, the floating side of the CMS swap will be set equal to the then-prevailing 3-year swap rate.

One thing we need to consider is how much of the hedge the dealer should put on. The net cash flow that will result from the CMS trade in a year is

$$\$100\text{mm} \times (s - 5\%)\,, \tag{6.13}$$

where s denotes the 3-year CMS setting in a year. As mentioned previously, for every bp change in 3-year CMS at maturity, the net cash will change by \$10,000. Thus, the dealer will not want to do \$100mm of the 1x4 swap to hedge. That would be too much. Suppose instead that the dealer receives on a notional of

$$\frac{\$100\text{mm}}{\frac{1}{(1.05)^1} + \frac{1}{(1.05)^2} + \frac{1}{(1.05)^3}} = \$36{,}720{,}856. \tag{6.14}$$

Scaling the notional down in this way adjusts for the fact that a 1 bp change in the 3-year swap rate at maturity changes the value of the hedge by an amount equal to the present value of a 1 bp annuity over three years. Note that we are choosing the notional of the *forward starting* 1x4 swap based on the PV01 of the *spot* 3-year swap.

Suppose that the dealer holds the hedge until its effective date in one year and at that time unwinds the swap. In a year, the hedge will be a spot starting 3-year swap with a coupon of 5%. From Section 3.3.1 we know that the value of this swap will be equal to the present value of an annuity of the difference between the coupon on the swap (which is 5%) and the then-prevailing 3-year swap rate, which we denote by s. That is, in one year the value of the hedge will be

$$\$36{,}720{,}856 \times \left\{ \frac{5\% - s}{[1+\mathrm{zc}(1)]^1} + \frac{5\% - s}{[1+\mathrm{zc}(2)]^2} + \frac{5\% - s}{[1+\mathrm{zc}(3)]^3} \right\}, \tag{6.15}$$

where zc(1), zc(2), and, zc(3) represent the then-prevailing zero coupon rates for one, two, and three years, respectively.

Under the assumption that the curve will move only in parallel shifts and remain flat, these zero coupon rates that will prevail in a year will all be equal to one another and furthermore will be equal to the 3-year swap rate prevailing in a year. Thus, recognizing that $\mathrm{zc}(1) = \mathrm{zc}(2) = \mathrm{zc}(3) = s$, we can rewrite Equation 6.15 in terms of s. The value of the hedge in one year is

$$\$36{,}720{,}856 \times \left[\frac{5\% - s}{(1+s)^1} + \frac{5\% - s}{(1+s)^2} + \frac{5\% - s}{(1+s)^3} \right]. \tag{6.16}$$

So the P&L in a year from the dealer's perspective resulting from the CMS trade and the hedge collectively are equal to the sum of the quantity in Equation 6.13 and the quantity in Equation 6.16:

$$\begin{aligned} &\$100\text{mm} \times (s - 5\%) \\ &\quad + \$36{,}720{,}856 \times \left[\frac{5\% - s}{(1+s)^1} + \frac{5\% - s}{(1+s)^2} + \frac{5\% - s}{(1+s)^3} \right]. \end{aligned} \tag{6.17}$$

This future P&L depends on one unknown variable: the prevailing 3-year swap rate in a year. Although this quantity is uncertain as of today, we can graph it as a function of the 3-year swap rate prevailing in a year. This is shown in Figure 6.22. The interesting thing to note here is that the dealer's P&L is always nonnegative and is strictly positive except when the 3-year swap rate in a year happens to be 5%.

But this then would be an arbitrage. Think about it: the dealer entered into the CMS swap and the hedge with no upfront cash flows, held both trades for a year, and had nonnegative cash flows in a year (and strictly positive cash flows, provided the 3-year swap rate in a year is not 5%). Since we preclude arbitrage opportunities, this must mean that our assumption that the fixed rate in the CMS swap is 5% must be wrong. In fact, the fixed rate in the CMS swap must be above 5%; since the dealer is paying the fixed rate, moving the fixed rate above 5% will mean that there are some scenarios in which the dealer will have negative P&L on the trade in a year, thereby precluding arbitrage.

It seemed so reasonable that the fixed rate in the CMS swap should be 5%. What's going on here? Look back at Equation 6.13, which shows the net cash

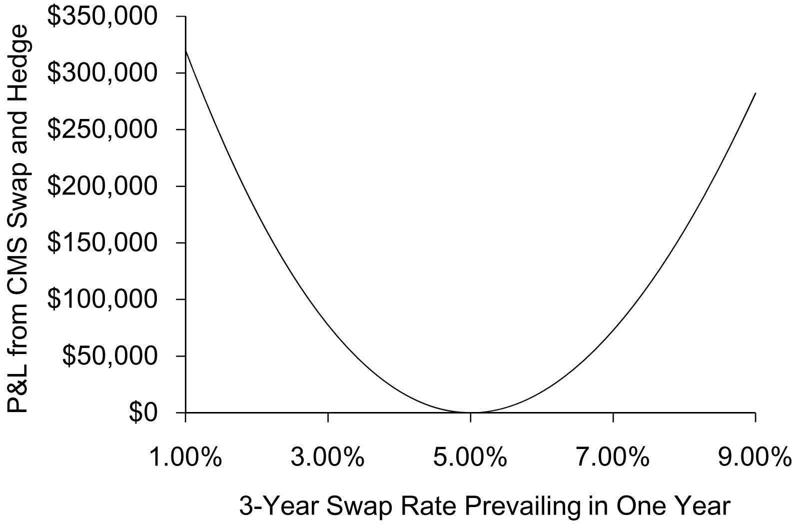

Figure 6.22: Dealer P&L in One Year from CMS Swap and Hedge

flows in the CMS swap in a year from the point of view of the dealer. The dealer is receiving s, the 3-year swap rate, and the dependence is linear in nature. But now look back at Equation 6.16. This represented the value of the 1x4 swap the dealer put on as a hedge as a function of the same s, the 3-year swap rate in a year. But this value is a convex function of s. We chose the notional on the 1x4 hedge based on the PV01 of a spot 3-year swap. But as rates go down, the PV01 goes up. And as rates go up, the PV01 goes down. Much as the buyer of a fixed rate bond benefits from the convexity of the bond by gaining more than he loses given equal shifts in the yield curve, the dealer is benefiting similarly from moves in the 3-year swap rate when he receives in the 1x4.

The amount that the fixed rate in the CMS swap must be moved up above the 1x4 forward swap rate is due to the *convexity adjustment*. How large is the adjustment? That is beyond the scope of our discussion.[42] However, we can make a few observations. First, Figure 6.22 suggests that the more volatile the (forward) 3-year swap rate is, the greater the convexity adjustment. The dealer is long vol in this trade; an increase in vol increases the value of the trade to the dealer. Second, the magnitude of this convexity adjustment increases as the tenor of the underlying swap in the CMS swap increases.

[42] For a more in-depth development of pricing the convexity adjustment, see, for example, Brigo and Mercurio (2006), Pugachevsky (2001), and Hagan (2003).

That is, the P&L profile would have been greater if we had considered a CMS swap on, say, 10-year CMS instead of 3-year CMS. Much like the convexity of a fixed rate bond increases as the maturity of the bond increases, the value of the convexity increases with an increase in this tenor. Third, as the maturity of the trade increases (the trade in Figure 6.9 is a 1-year trade), the convexity adjustment increases as well.

The profile exhibited in Figure 6.22 is somewhat reminiscent of a *straddle*, a position consisting of a call option and a put option with the same strike. This suggests that in order to hedge his exposure to vol, the dealer would want to sell some options (perhaps, heuristically, some 1 into 3 receiver swaptions and 1 into 3 payer swaptions struck at 5%). Selling these options would help flatten out the P&L profile in Figure 6.22. The 1x4 hedge was put on by the dealer so as to hedge the delta of the CMS trade. The options would be sold to hedge the gamma of the CMS trade. The dealer who is essentially getting long these options must compensate the client for them: the increase in the fixed rate in the CMS swap due to the convexity adjustment can be interpreted, heuristically, as compensation for these options.

In this section we considered a simplified case in which the swap curve was flat and only moved in parallel shifts. Doing so allowed us to isolate the nature of the convexity adjustment to just one variable and display the nature of the adjustment easily as we did in Figure 6.22. The same forces are at work (meaning, convexity adjustments are applicable) if we consider any arbitrary curve shape and movements in the curve, but the analysis becomes more complicated.